# Comparative Analysis of Deep Learning Approaches for Crop Disease Multi-classification (CNN vs. Pre-trained EfficientNetB0 vs. Hybrid Model)

**B. Tech. Project Report**

*Submitted by*


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**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING,**

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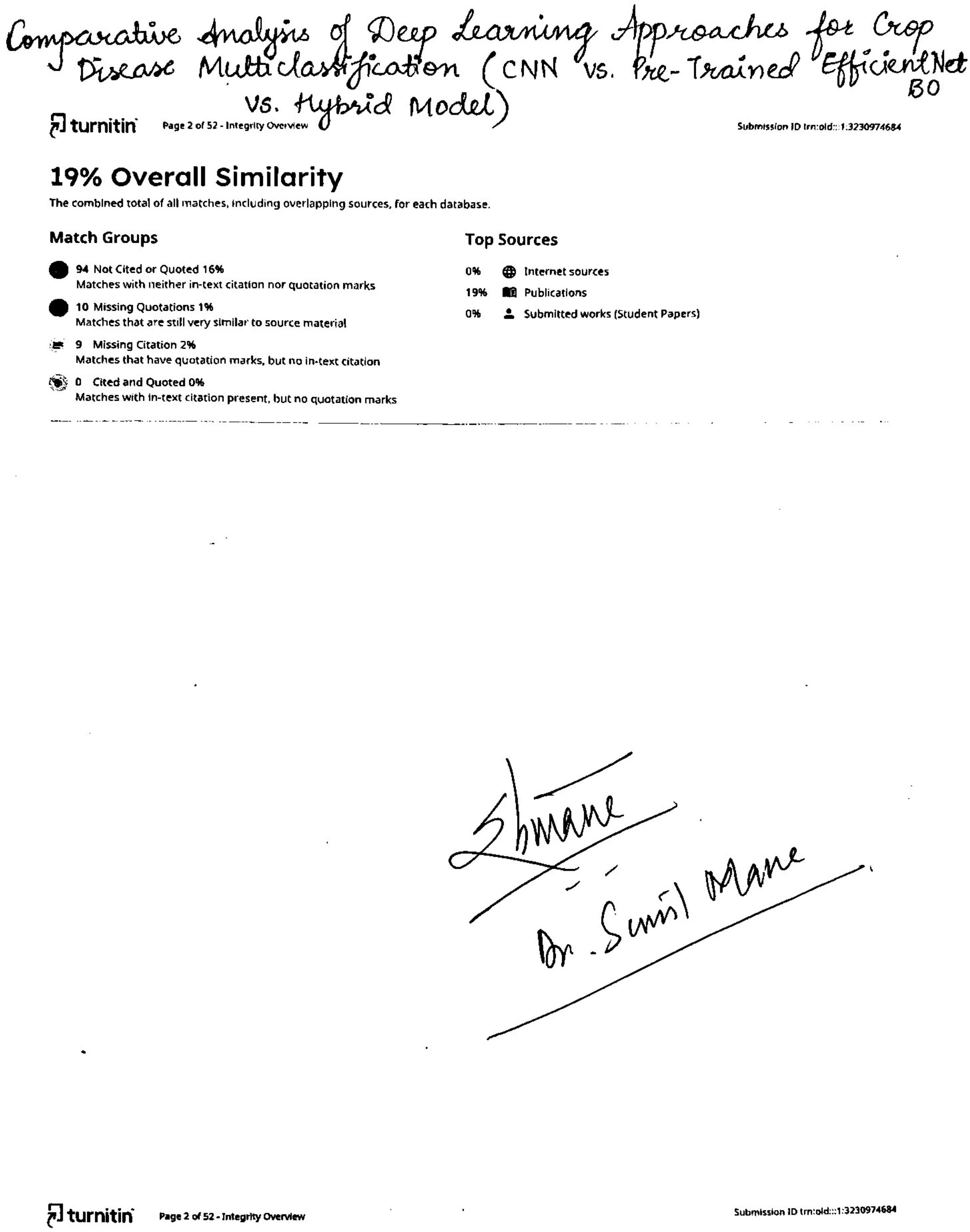
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Figure 1: Plagiarism report

# Abstract

This study presents a comparative evaluation of three deep learning models— Customized Convolutional Neural Network(CNN), pre-trained EfficientNetB0, and a Hybrid architecture combining both models—for the task of multi-class crop disease classification. Models were trained on two systematically balanced datasets: PlantVillage(pv) and New Plant Diseases(npd). Each model's performance was assessed using key classification metrics including accuracy, precision, recall, and F1-score.

Results showed that EfficientNetB0 marginally outperformed both the Customized CNN and the Hybrid Model in terms of accuracy on the New Plant Disease dataset, achieving 99.89%. On the PlantVillage dataset, the Hybrid Model achieved the highest test accuracy of 99.82%, closely followed by EfficientNetB0 with 99.80%. While the Customized CNN showed slightly lower performance overall, it still demonstrated robust classification capabilities, particularly on the more controlled PlantVillage dataset.

The findings suggest that while pre-trained models like EfficientNetB0 offer strong generalization and computational efficiency, hybrid approaches may be viable alternatives when required for fine-grained domain-specific feature extraction. This work provides insights into how dataset balancing and model architecture impact classification outcomes and offers practical directions for designing deep learning-based agricultural diagnostic tools.

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# Chapter 1

# Introduction

Plant diseases represent a critical challenge affecting agricultural productivity, crop quality, and global food security [8, 34]. Traditional diagnostic methods, which rely heavily on visual inspection and expert knowledge, are often labor-intensive, time-consuming, and prone to subjectivity and human error. These limitations highlight the need for automated and reliable plant disease detection systems capable of delivering rapid and accurate diagnoses to support timely intervention and minimize crop losses [7].

Recent advances in artificial intelligence—particularly deep learning techniques such as Convolutional Neural Networks (CNNs) and transfer learning—have proven highly effective for plant disease detection from image data [31, 27, 32]. Pre-trained models like EfficientNet have demonstrated outstanding performance in general-purpose image classification tasks and are increasingly being adapted for plant disease classification [12, 17]. However, while these models provide robust generalization, their ability to recognize subtle, domain-specific features may be limited. On the other hand, customized CNN architectures, tailored specifically for plant pathology, have shown the capacity to extract fine-grained disease-related features but often struggle with generalizability and computational efficiency [28, 4].

To bridge this gap, hybrid models are emerging that integrate the strengths of both custom CNNs and pre-trained architectures. Such models aim to harness the detailed feature extraction capabilities of domain-specific CNNs alongside the scalable and transferable benefits of architectures like EfficientNet [1].

In this context, our study presents a comparative analysis of three deep learning approaches: (1) a customized CNN designed specifically for plant disease detection, (2) a fine-tuned EfficientNetB0 model, and (3) a novel hybrid architecture that combines both methodologies. These models are trained and evaluated on two systematically balanced datasets—PlantVillage and New Plant Diseases—curated using targeted augmentation techniques to ensure equitable class representation. This research explores the effect of data balancing on model generalization and aims to determine whether hybrid architectures yield enhanced performance across key metrics.

The significance of this work lies in its contribution to the understanding of how dataset design and model architecture interact to affect classification performance. The results provide insights into the trade-offs between accuracy, computational complexity, and generalizability, ultimately informing the development of scalable, real-world solutions for precision agriculture and food security.

# Chapter 2

# Literature Review

Recent advancements in deep learning have significantly enhanced plant disease detection and classification, particularly with the integration of convolutional neural networks (CNNs) and transfer learning (TL) techniques. This section discusses key studies relevant to the field and contextualizes our work.

## 2.1 Impact of Class Imbalance on CNNs

Buda et al. [10] systematically studied the effect of class imbalance on CNN performance. They demonstrated that oversampling methods can significantly improve classification accuracy without causing overfitting, thereby highlighting the necessity of dataset balancing strategies.

## 2.2 Interpretability in Plant Disease Detection

Brahimi et al. [9] proposed a Teacher-Student CNN architecture to enhance interpretability for plant disease classification. Their work, tested on the PlantVillage dataset, showcased sharper visualization of important image regions, improving model transparency.

## 2.3 Hybrid Deep Learning Approaches

Altalak et al. [5] presented a hybrid model combining CNN, CBAM, and SVM for tomato leaf disease detection. Achieving an accuracy of 97.2%, their lightweight model demonstrated suitability for deployment on resource-constrained smart devices.

Aboelenin et al. [1] proposed a hybrid framework integrating CNNs (VGG16, InceptionV3, DenseNet201) with Vision Transformers (ViT), achieving high classification accuracy (99.24% for apple, 98% for corn). Their architecture effectively captured both global and local features.

## 2.4 Explainable Deep Learning Models

Nigar et al. [22] developed an Explainable AI system combining EfficientNetB0 and LIME, achieving a classification accuracy of 99.69%. Their approach enhances model transparency and supports better decision-making in agricultural settings.

## 2.5 Ensemble and Optimized CNN Models

Muthusamy et al. [21] proposed the IncepV3Dense ensemble model combining InceptionV3 and DenseNet169, achieving 98.62% validation accuracy for detecting micronutrient deficiencies in banana crops.

Sohail et al. [33] developed a metaheuristic-optimized hybrid CNN-LBP model, achieving 99.8% accuracy for plant leaf disease classification, showcasing the potential of ensemble learning combined with optimization techniques.

## 2.6 Mobile Applications and Lightweight Models

Salam et al. [30] introduced a modified MobileNetV3Small CNN model for mulberry leaf disease detection, achieving 96.4% accuracy. Their solution, designed as a mobile application, enables real-time disease diagnosis for sericulture farmers.

## 2.7 EfficientNet Variants for Disease Detection

Ahmad et al. [2] leveraged Efficient CNNs with stepwise transfer learning to tackle plant disease detection in imbalanced datasets, achieving 99.69% accuracy on PlantVillage and 99% on pepper datasets.

Sunil et al. [14] proposed an EfficientNetV2-based approach for cardamom plant disease detection, achieving a high detection accuracy of 98.26%, emphasizing its applicability in real-time agricultural solutions.

## 2.8 Comparative Studies and Hybrid Models

Altabaji et al. [11] performed a comparative analysis of LeafNet, Modified LeafNet, MobileNetV2, and Xception for rice leaf disease classification. Modified LeafNet achieved the highest validation accuracy of 97.44%.

Alam et al. [3] compared CNN, KNN, and a hybrid EfficientNetB0 model for potato disease classification. Their hybrid model achieved a superior accuracy of 99.77%, significantly outperforming the other approaches.

## 2.9 Summary

The reviewed studies highlight that while transfer learning and pre-trained architectures offer high accuracy, hybrid and ensemble strategies further enhance classification performance. Interpretability, efficient deployment, and addressing class imbalance are critical factors for practical agricultural applications.

# Chapter 3

# Research Gaps and Problem Statement

Despite substantial progress in deep learning-based crop disease classification, several gaps remain unaddressed in existing literature. This chapter outlines the major research gaps and formally states the problem investigated in this project.

## 3.1 Research Gaps

### 3.1.1 Handling Class Imbalance Effectively

Many existing studies such as Buda et al. [10] demonstrate that class imbalance significantly impacts model performance. However, several recent works continue to use imbalanced datasets without appropriate rebalancing, leading to biased classification outcomes.

### 3.1.2 Generalization Across Diverse Crop Diseases

Pre-trained models like EfficientNetB0 have achieved strong performance on popular datasets (e.g., PlantVillage), but their ability to generalize to more challenging, real-world datasets with subtle inter-class variations remains less explored.

### 3.1.3 Lack of Comprehensive Comparative Studies

Few studies systematically compare customized CNNs, pre-trained models, and hybrid architectures under the same experimental conditions across balanced datasets. Ablation studies that assess performance trade-offs between model types remain scarce.

### 3.1.4 Interpretability and Practical Deployment Challenges

While recent works attempt to integrate explainability tools like LIME and Grad-CAM [22], many models are still treated as black boxes. Furthermore, mobile deployment suitability for farmers, especially with hybrid or ensemble models, is an underexplored area.

## 3.2 Problem Statement

Current plant disease classification systems often rely on either customized CNNs, which may capture domain-specific features but suffer from limited scalability, or pre-trained models like EfficientNetB0, which generalize well but might miss fine-grained details critical for accurate disease identification.

Moreover, the majority of studies focus on isolated models rather than examining hybrid approaches that could potentially offer an optimal trade-off between accuracy, computational efficiency, and robustness.

Thus, the primary problem addressed in this project is:

> *To comparatively evaluate a Customized CNN, a Pre-trained EfficientNetB0, and a Hybrid Model combining both approaches for multi-class crop disease classification on systematically balanced datasets,*

*analyzing their performance in terms of accuracy, precision, recall, F1-score, generalization capability, and computational cost.*

## 3.3 Objectives

The specific objectives of this research are:

- To develop a Customized CNN model tailored for crop disease multi-classification.
- To fine-tune a pre-trained EfficientNetB0 model on balanced datasets.
- To construct a hybrid model combining features from both architectures.
- To perform systematic evaluation comparing these approaches.
- To assess each model based on classification metrics and training efficiency.

## 3.4 Scope of the Study

This study focuses on two curated datasets: the PlantVillage (PV) dataset and the New Plant Diseases (NPD) dataset, both balanced across multiple classes. The results are intended to provide practical guidelines for selecting suitable deep learning models for agricultural disease diagnosis systems.

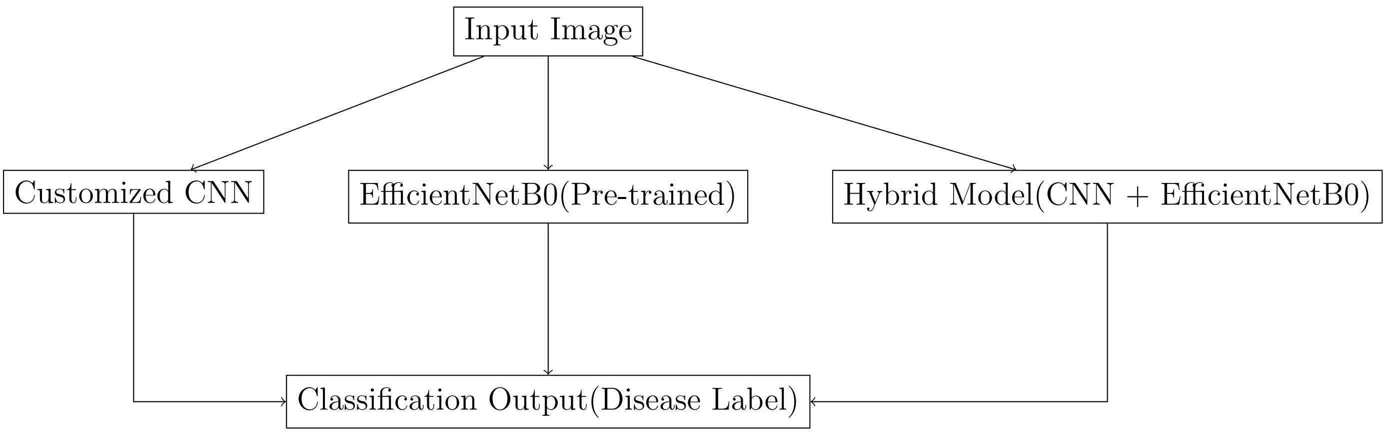


Figure 3.1: Conceptual overview of the model comparison approach adopted in this study.

# Chapter 4

# Proposed Methodology/ Solution

## 4.1 Materials

This study utilized publicly available datasets, computational resources, and software libraries tailored for deep learning and data processing tasks. The hardware configuration included an NVIDIA GeForce RTX 3050 GPU (4 GB VRAM), a multi-core processor, and 16 GB RAM. The software environment comprised Python 3.8, the PyTorch deep learning framework, and associated libraries such as `torchvision`, `scikit-learn`, `NumPy`, `pandas`, and `Matplotlib` for data handling, model development, and result visualization. Additional utilities included `tqdm` for progress monitoring during training, `albumentations` for advanced data augmentation, and built-in Python modules like `os` and `glob` for file and directory management.

## 4.2 Data Collection and Dataset Description

Two plant disease image datasets were selected to evaluate model performance comprehensively: the *PlantVillage Dataset* and the *New Plant Diseases Dataset*. The PlantVillage dataset initially contained 61,486 images across 39 classes, representing various healthy and diseased plant conditions under controlled laboratory settings [8]. After balancing, each class was stan-

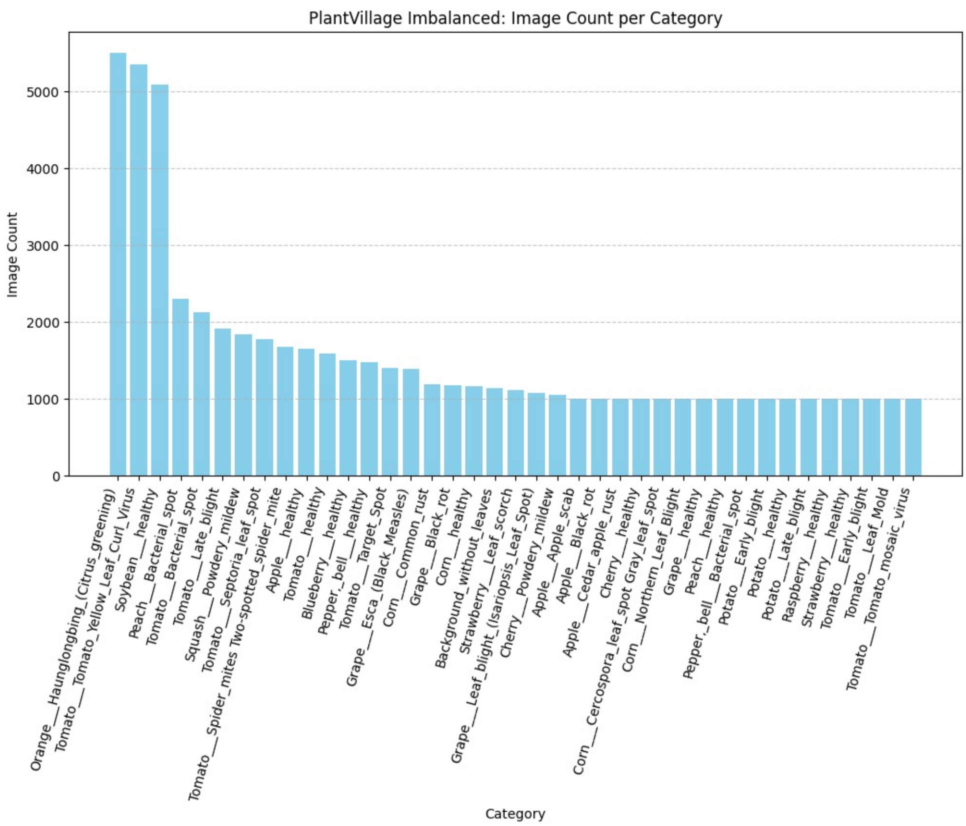


Figure 4.1: Class distribution of the PlantVillage dataset before balancing

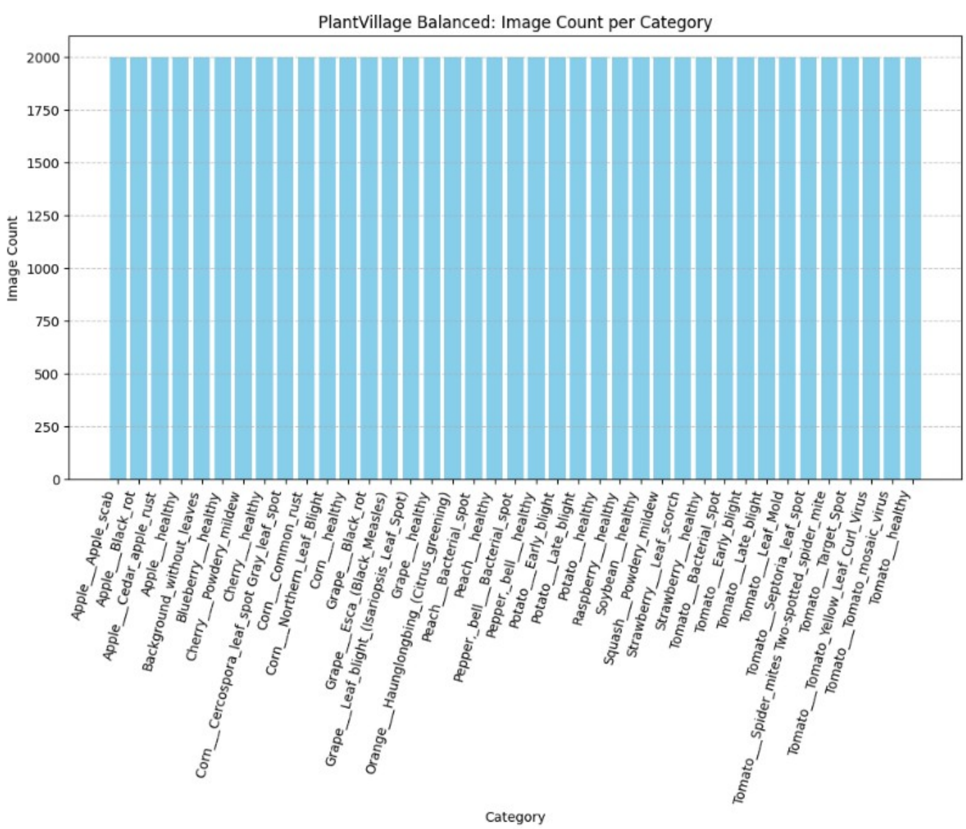


Figure 4.2: Class distribution of the PlantVillage dataset after balancing

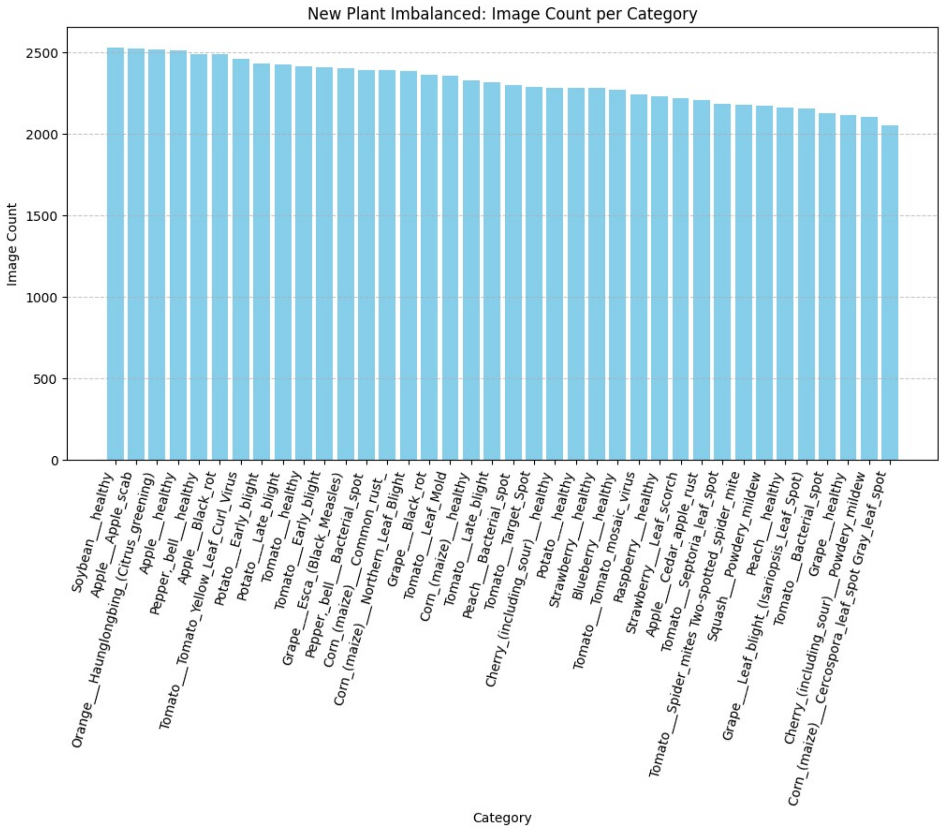


Figure 4.3: Class distribution of the New Plant Diseases dataset before balancing

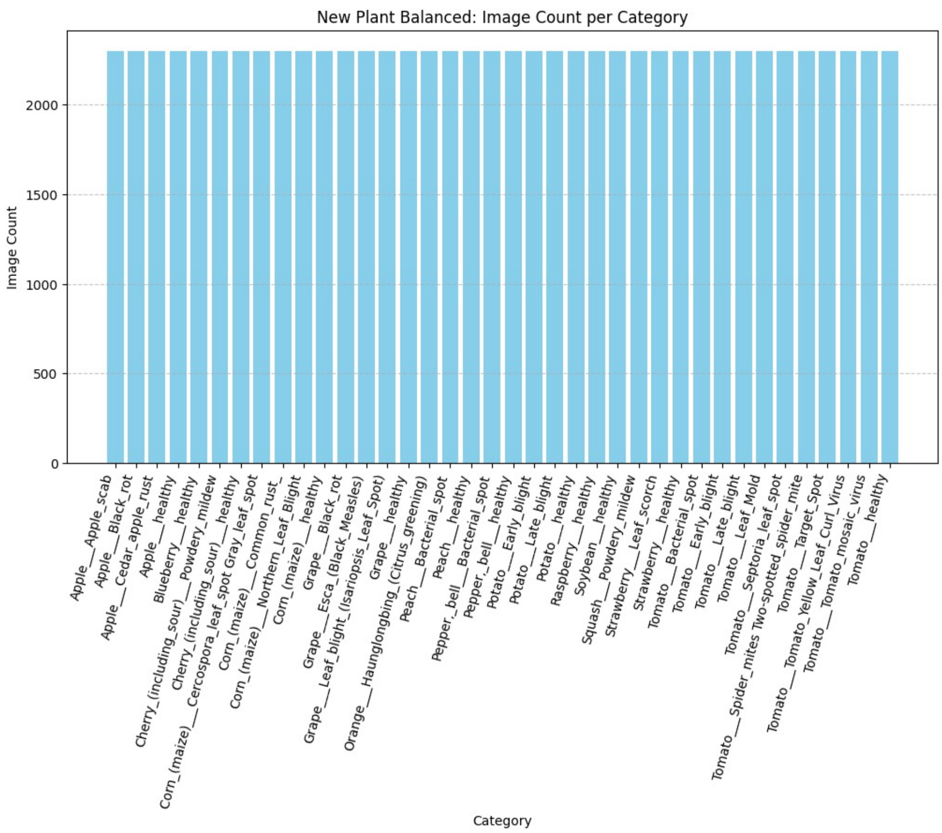


Figure 4.4: Class distribution of the New Plant Diseases dataset after balancing

dardized to 2,000 images, resulting in 78,000 images.

The New Plant Diseases dataset, comprising more realistic images, originally included 87,900 images spanning 38 classes [31]. Balancing procedures standardized each category to approximately 2,300 images per class, yielding a balanced dataset of 87,400 images. Balancing was achieved through data augmentation techniques, including rotation, flipping, zooming, contrast adjustments, strategic oversampling, and undersampling, ensuring equal class representation [29].

Fig. 4.1 and Fig. 4.2 illustrate the class distribution of the PlantVillage dataset before and after balancing, respectively. Similarly, Fig. 4.3 and Fig. 4.4 depict the class distribution of the New Plant Diseases dataset before and after applying balancing techniques. These visualizations clearly demonstrate the effectiveness of the dataset balancing procedures in achieving uniform class distributions across categories.

# Chapter 5

# Experimental Setup

Three distinct deep learning approaches were comparatively analyzed:

A *Customized CNN* architecture was designed to capture domain-specific visual features from plant leaf images. It consisted of four convolutional blocks, each including two Conv2D layers with ReLU activation and batch normalization, followed by max-pooling for spatial downsampling. Fully connected layers included a dense layer with 1024 neurons and a dropout of 40% to reduce overfitting, ending with a softmax classification layer.

A *Pre-trained EfficientNetB0 model* utilized transfer learning by fine-tuning weights initially trained on ImageNet Dataset (contains 14,197,122 annotated images according to the WordNet hierarchy). The classification head was replaced to match plant disease categories, enabling efficient learning. EfficientNet's compound scaling allows optimization of accuracy and computational load [17].

A *Hybrid Model* was constructed by integrating features from both the customized CNN and EfficientNetB0. Independent feature extraction was followed by concatenation, and the merged vector was passed through fully connected layers for classification. This fusion approach aims to combine the domain specificity of CNNs with the generalization capability of pre-trained models [5, 1].

## 5.1 Computational Methods and Training Details

All models were implemented using PyTorch. Each model underwent training for a maximum of 50 epochs (early stop was used) with a batch size of 16. Cross-entropy loss served as the loss function, and the Adam optimizer with adaptive learning rates and weight decay was employed to optimize the model parameters. Regularization techniques, including dropout and batch normalization, were consistently applied across experiments to reduce overfitting [25].

To ensure unbiased model evaluation, stratified hold-on sampling was employed to split the dataset into approximately equal proportions for training, validation, and testing. The learning rate was reduced from 0.001 to 0.0001, and the early stopping delta from 0.001 to 0.0001, to enable more stable fine-tuning and allow the model to continue training despite small but meaningful improvements in validation performance [13]. Additionally, a Cosine Annealing Learning Rate Scheduler was used in place of ReduceLROnPlateau, as it provides a smooth, periodic decay in learning rate that encourages better exploration of the loss landscape and often leads to improved generalization compared to abrupt reductions based on validation plateaus.

Additionally, an early stopping criterion was implemented to monitor the validation loss, terminating training if no improvement was observed for 5 consecutive epochs, thereby preventing overfitting and ensuring optimal generalization. Dynamic learning rate scheduling further enhanced training efficiency and convergence. Model performance was monitored through accuracy, precision, recall, F1-score, and confusion matrices, which are standard metrics in classification tasks [11].

## 5.2 Statistical and Computational Evaluation Techniques

Model evaluation involved detailed statistical analyses of classification metrics, primarily accuracy, precision, recall, and F1-score [11]. Accuracy was computed as the ratio of correctly classified samples to the total number of samples, as shown below:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{5.1}$$

Precision, representing the proportion of correctly predicted positive observations relative to the total predicted positives, was computed as shown in Equation 5.2. Recall, which measures the proportion of correctly identified positives relative to actual positives, was computed as shown in Equation 5.3.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5.2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{5.3}$$

Here, $TP$ (True Positive) represents the correctly predicted positive observations, $TN$ (True Negative) denotes correctly predicted negative observations, $FP$ (False Positive) refers to incorrectly predicted positive observations, and $FN$ (False Negative) indicates incorrectly predicted negative observations. The F1-score, an indicator of balanced precision and recall, was computed as their harmonic mean.

Confusion matrices were generated to visually assess misclassification patterns and class-wise performance [27]. Computational efficiency was evaluated based on training time, convergence rates, GPU memory consumption, and inference speed. Results were systematically recorded and visualized to facilitate comparative performance analyses across the three modeling approaches and dataset balancing strategies.

# Chapter 6

# Results and Discussion

## 6.1 Results

### 6.1.1 Customized CNN

#### Plant Village Dataset

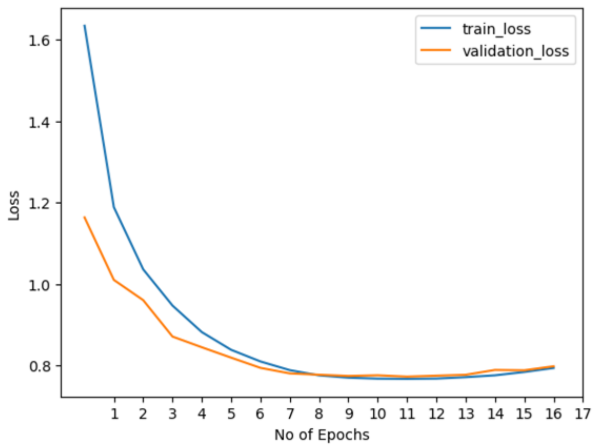


Figure 6.1: Training and validation loss on the PlantVillage dataset using customized CNN

This CNN model achieved an overall accuracy of 98.24% on the test dataset, demonstrating robust performance across various plant disease categories. The lowest category accuracy was 87.5% for "Tomato—Early Blight," likely resulting from symptom similarity with diseases such as Late Blight, Leaf Mold, and Bacterial Spot, which caused misclassification.

The overall precision, recall, and F1-score were approximately 98.25%, 98.24%, and 98.23%, respectively, as shown in Table 6.1, indicating an effective balance between false positives and false negatives. This performance suggests successful data augmentation and class-balancing strategies that mitigated imbalance issues.

Loss curves for training and validation sets exhibited rapid convergence within ten epochs (Figure 6.1), stabilizing thereafter, leading to early stopping at epoch 17. This behavior indicates optimal learning without significant overfitting. Each epoch took approximately 9 minutes, likely influenced by the simplicity of the CNN architecture and input data resolution.

The confusion matrix (Figure 6.2) showed strong diagonal dominance, with misclassifications mainly among visually similar tomato diseases. This highlights the challenges posed by symptom overlaps, making it challenging for the CNN model to distinguish between them.

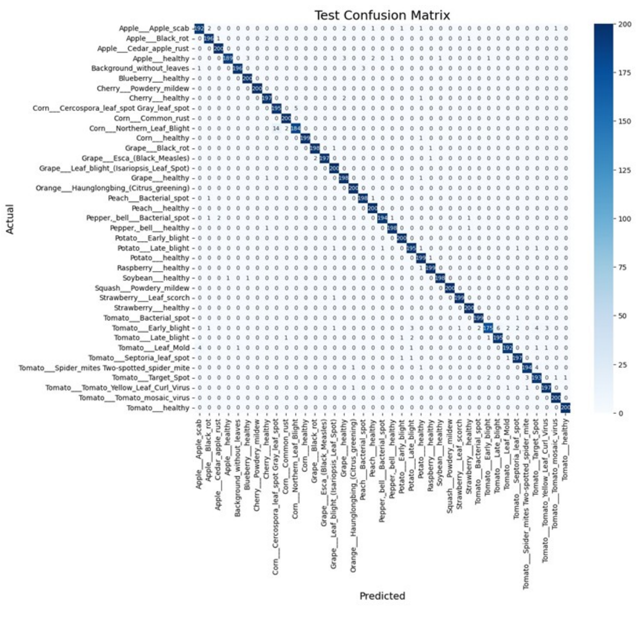


Figure 6.2: Confusion matrix of PlantVillage dataset using customized CNN

Table 6.1: Performance of customized CNN on the PlantVillage dataset

| Dataset | Accuracy | Precision | Recall | F1-score |
|---|---|---|---|---|
| Validation | 98.5% | 98.51% | 98.50% | 98.50% |
| Test | 98.24% | 98.25% | 98.24% | 98.23% |

**New Plant Diseases Dataset**

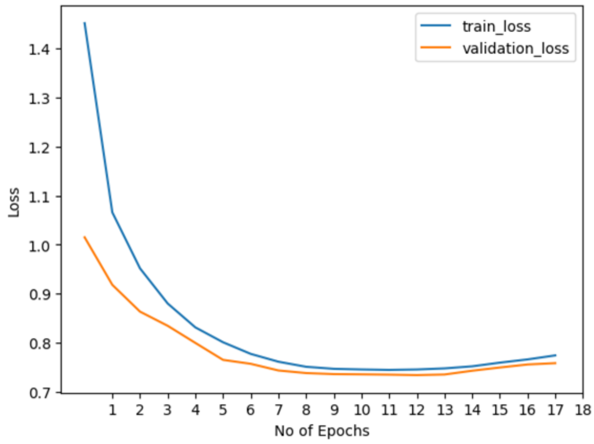


Figure 6.3: Training and validation loss on the New Plant Diseases dataset using customized CNN

The CNN model attained an overall accuracy of 99.16% on the test dataset. The lowest per-category accuracy was 93.04% for "Corn (maize)—Cercospora Leaf Spot Gray Leaf Spot," attributed to symptom overlap with diseases such as Northern Leaf Blight and Common Rust.

Precision, recall, and F1-score were around 99.17%, 99.16%, and 99.16%, respectively (Table 6.2), reflecting a well-balanced classification capability. Adequate dataset preparation and augmentation contributed significantly to this high performance.

Loss curves indicated rapid initial convergence and stabilized after approximately ten epochs, triggering early stopping at epoch 18 (Figure 6.3). This suggests that the model reached an optimal balance between underfitting and overfitting, as subsequent epochs showed negligible improvement. Each epoch took approximately 10 minutes, likely due to the larger dataset size than the Plant Village dataset.

The confusion matrix (Figure 6.4) highlighted accurate predictions across most categories, with misclassification primarily within visually similar maize and tomato diseases.

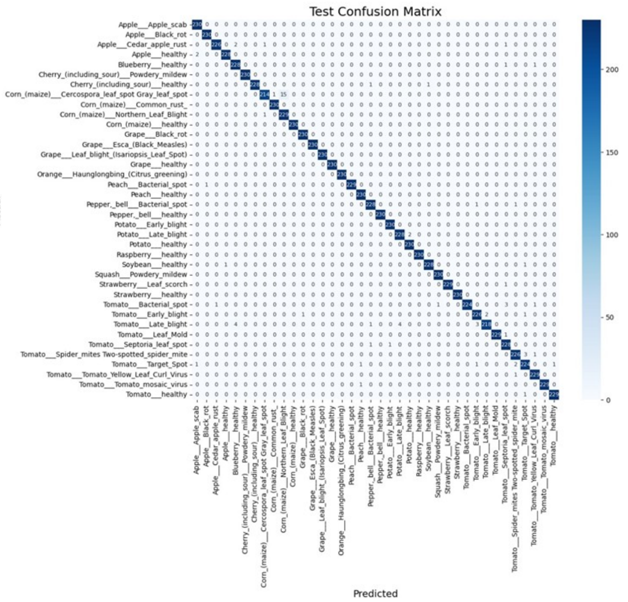


Figure 6.4: Confusion matrix of the New Plant Diseases dataset using customized CNN

Table 6.2: Performance of customized CNN on the New Plant Diseases dataset

| Dataset | Accuracy | Precision | Recall | F1-score |
|---|---|---|---|---|
| Validation | 99.08% | 99.09% | 99.09% | 99.08% |
| Test | 99.16% | 99.18% | 99.16% | 99.16% |

### 6.1.2 Pre-trained EfficientNetB0

#### Plant Village Dataset

The EfficientNetB0 model achieved an overall accuracy of 99.79%. The lowest per-class accuracy was 97% for "Corn—Northern Leaf Blight," likely due to subtle symptom distinctions from other maize leaf diseases.

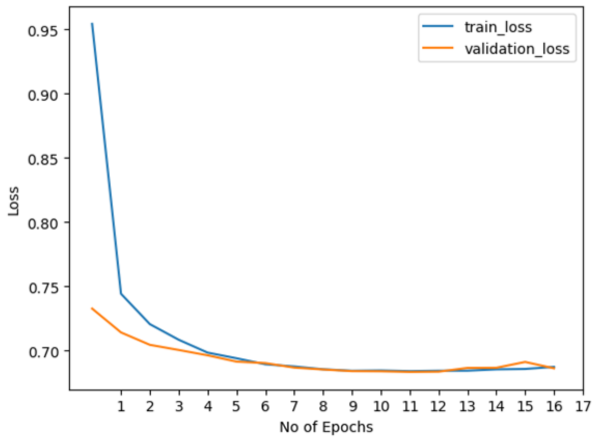


Figure 6.5: Training and validation loss on the PlantVillage dataset using EfficientNetB0

Precision, recall, and F1-score were approximately 99.80%, 99.79%, and 99.79%, respectively (Table 6.3), reflecting the model's effective transfer learning capabilities.

Training and validation losses converged rapidly within initial epochs, stabilizing early and leading to early stopping at epoch 17 (Figure 6.5). Each epoch required approximately 8 minutes, benefiting from EfficientNet's optimized structure and pre-training.

The confusion matrix (Figure 6.6) demonstrated clear diagonal dominance, with minimal misclassifications among visually similar maize disease categories.

Table 6.3: Performance of EfficientNetB0 on the PlantVillage dataset

| **Dataset** | **Accuracy** | **Precision** | **Recall** | **F1-score** |
|---|---|---|---|---|
| Validation | 99.83% | 99.83% | 99.83% | 99.83% |
| Test | 99.80% | 99.80% | 99.80% | 99.80% |

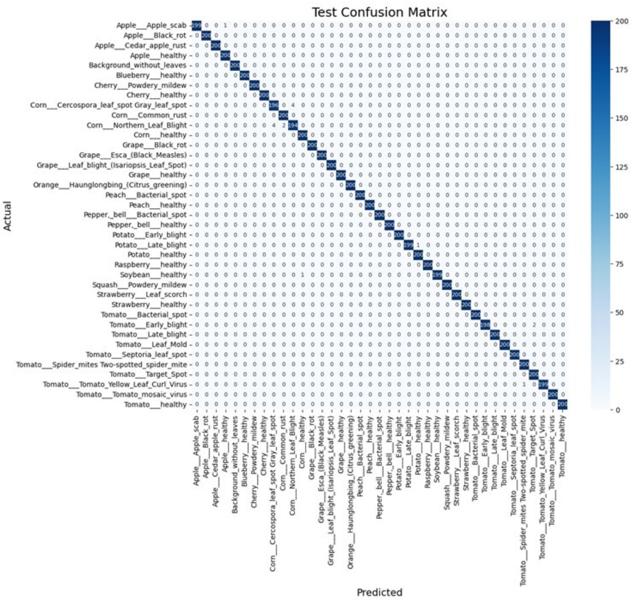


Figure 6.6: Confusion matrix of the PlantVillage dataset using EfficientNetB0

**New Plant Diseases Dataset**

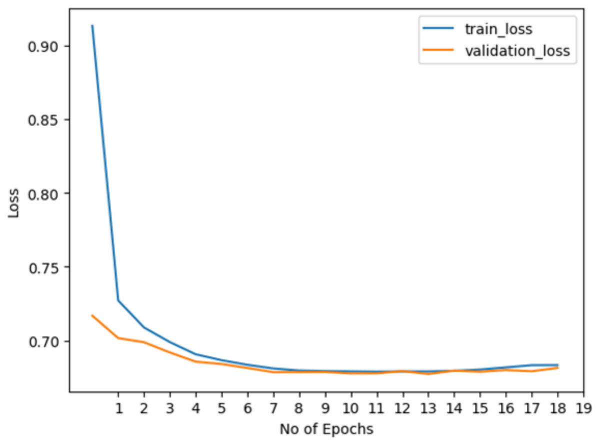


Figure 6.7: Training and validation loss on the New Plant Diseases dataset using EfficientNetB0

EfficientNetB0 recorded an overall accuracy of 99.89%. The category "Corn (maize)—Cercospora Leaf Spot Gray Leaf Spot" had the lowest accuracy (96.96%) due to its visual similarities with other maize diseases.

Overall precision, recall, and F1-score values were around 99.89% (Table 6.4), indicating strong performance in disease classification. Efficient feature extraction from transfer learning was a major contributing factor.

Loss curves showed rapid convergence and early stabilization, triggering early stopping at epoch 19 (Figure 6.7). Each epoch duration was about 7 minutes, attributable to EfficientNet's efficient computation and architecture.

The confusion matrix (Figure 6.8) highlighted accurate classification, with minor misclassifications predominantly among closely related maize diseases.

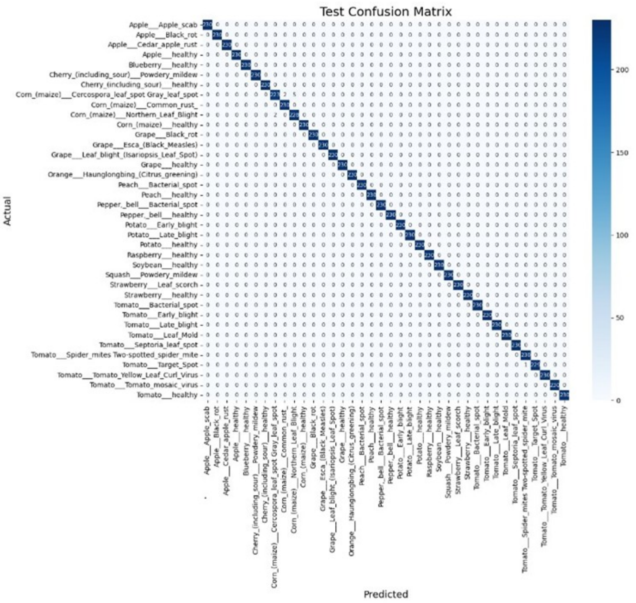


Figure 6.8: Confusion matrix of the New Plant Diseases dataset using EfficientNetB0

Table 6.4: Performance of EfficientNetB0 on the New Plant Diseases dataset

| **Dataset** | **Accuracy** | **Precision** | **Recall** | **F1-score** |
|---|---|---|---|---|
| Validation | 99.83% | 99.83% | 99.82% | 99.83% |
| Test | 99.89% | 99.89% | 99.89% | 99.89% |

### 6.1.3 Hybrid Model(Customized CNN + EfficientNetB0)

**Plant Village Dataset**

The Hybrid model achieved an overall test accuracy of 99.82%. "Corn—Northern Leaf Blight" recorded the lowest accuracy (96.5%), likely due to its symptom similarities with other maize diseases.

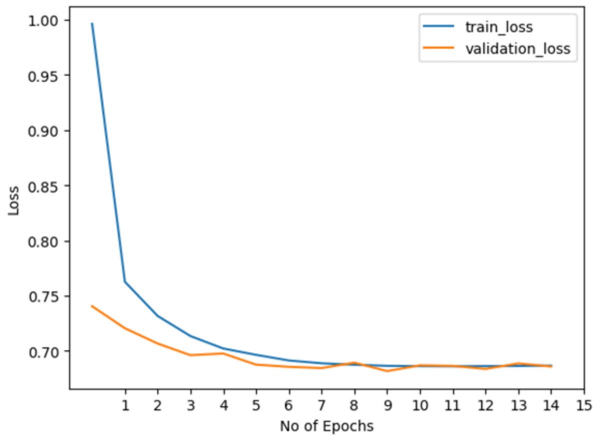


Figure 6.9: Training and validation loss on the PlantVillage dataset using the Hybrid Model

Precision, recall, and F1-score were approximately 99.82% each, highlighting the model's robust discriminative ability, which leverages customized CNN layers and EfficientNetB0's pre-trained weights (Table 6.5).

Loss curves rapidly converged within initial epochs, stabilizing by epoch 15 and triggering early stopping (Figure 6.9). Each epoch took around 34 minutes, significantly longer due to the combined complexity of the two architectures and increased computational demands.

The confusion matrix (Figure 6.10) indicated strong diagonal dominance, with minor misclassifications focused on visually similar maize diseases.

Table 6.5: Performance of the Hybrid Model on the PlantVillage dataset

| **Dataset** | **Accuracy** | **Precision** | **Recall** | **F1-score** |
|---|---|---|---|---|
| Validation | 99.72% | 99.72% | 99.72% | 99.72% |
| Test | 99.82% | 99.82% | 99.82% | 99.82% |

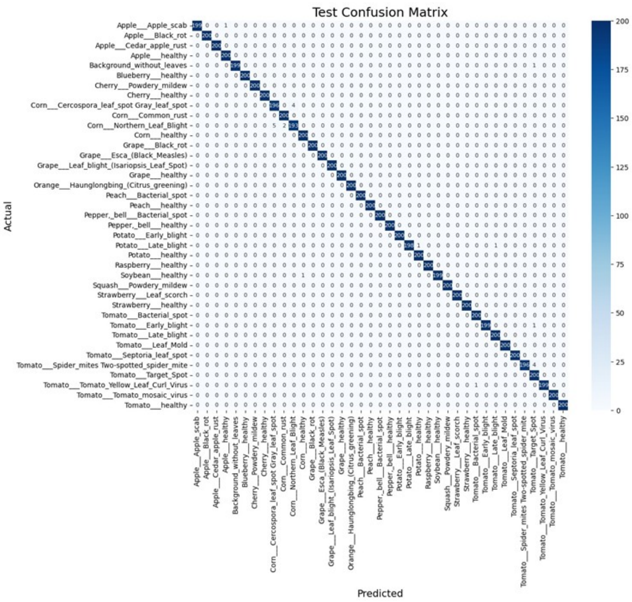


Figure 6.10: Confusion matrix of the PlantVillage dataset using the Hybrid Model

**New Plant Diseases Dataset**

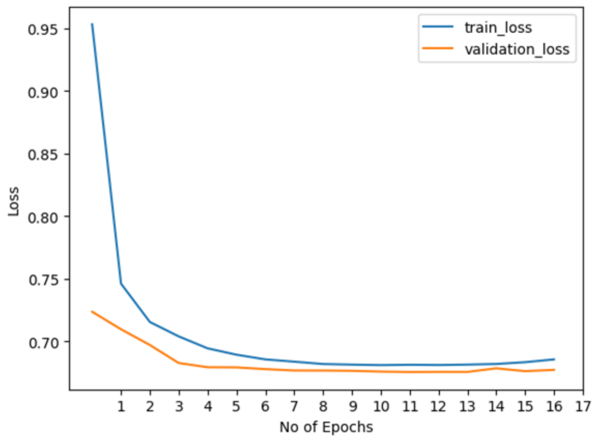


Figure 6.11: Training and validation loss on the New Plant Diseases dataset using the Hybrid Model

The Hybrid model achieved an overall accuracy of 99.87%. The lowest accuracy per category was 98.7% for "Tomato—Septoria Leaf Spot," attributed to visual similarities with other tomato diseases.

Precision, recall, and F1-score were approximately 99.88%, 99.87%, and 99.87%, respectively (Table 6.6), reflecting the effective integration of customized CNN architecture with EfficientNet's transfer learning.

Training and validation loss curves stabilized quickly within initial epochs, triggering early stopping at epoch 17 (Figure 6.11). Each epoch duration averaged around 34 minutes, caused by increased computational complexity from combining both CNN and EfficientNet architectures.

The confusion matrix (Figure 6.12) demonstrated high predictive reliability, with minimal misclassifications involving visually similar tomato diseases.

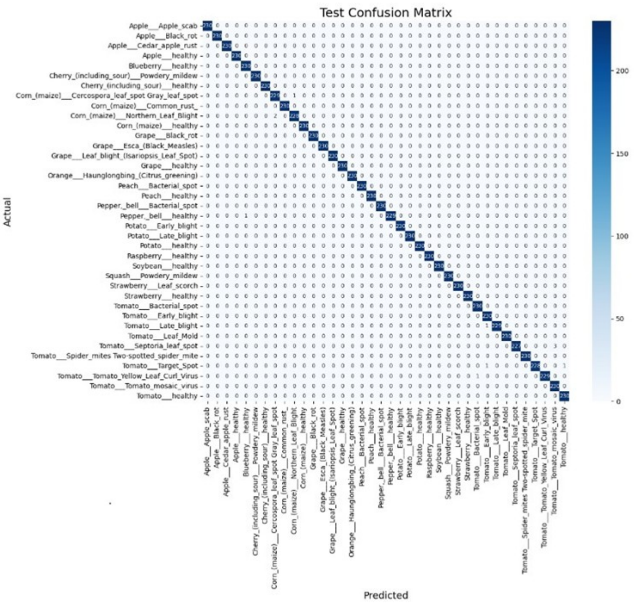


Figure 6.12: Confusion matrix of the New Plant Diseases dataset using the Hybrid Model

Table 6.6: Performance of the Hybrid Model on the New Plant Diseases dataset

| **Dataset** | **Accuracy** | **Precision** | **Recall** | **F1-score** |
|---|---|---|---|---|
| Validation | 99.87% | 99.88% | 99.87% | 99.87% |
| Test | 99.87% | 99.88% | 99.87% | 99.87% |

## 6.2 Discussion

### 6.2.1 Interpretation of Results

This study investigated the performance of three deep learning architectures across two balanced plant disease datasets. The Customized CNN showed competent performance, especially on the PlantVillage dataset, indicating its effectiveness in controlled environments [8, 25]. On the other hand, the pre-trained EfficientNetB0 consistently achieved the highest accuracy scores on both datasets, demonstrating its ability to generalize well even with complex and diverse field images [12, 17].

Although the Hybrid Model was hypothesized to combine the strengths of both architectures, its performance did not universally exceed that of EfficientNetB0. On the PlantVillage dataset, the Hybrid model performed comparably to EfficientNetB0, achieving 99.72% accuracy versus 99.80%. However, on the New Plant Diseases dataset, EfficientNetB0 slightly outperformed the Hybrid model (99.89% vs. 99.87%). These results suggest that while hybridization offers potential benefits, its increased complexity does not always translate to superior accuracy [1].

### 6.2.2 Implications and Practical Significance

Although this study was conducted on benchmarked datasets in a controlled setting, the results offer practical insights. The EfficientNetB0 model, due to its strong accuracy and lightweight architecture, is a promising candidate for deployment in mobile or edge-based agricultural applications. Meanwhile, the Hybrid Model, while computationally heavier, could be refined further for specialized scenarios requiring more nuanced disease feature extraction.

It is important to note that the models were not tested in real-world conditions, and as such, the performance metrics should not be interpreted as immediately transferable to live agricultural deployments. However, this study provides a foundational benchmark for future development and testing of AI-driven plant disease classification systems under realistic conditions.

### 6.2.3 Comparative Analysis with Prior Research

Table 6.7: Comparison of recent deep learning models for crop disease classification.

| **Paper & Citation** | **Objective** | **Model Type** | **Dataset(s)** | **Accuracy and Precision/Recall/F1** | **Key Techniques** |
|---|---|---|---|---|---|
| Improving Plant Disease Classification [22] | Interpretability in disease detection using DL | VGG16 + XAI | Balanced Plant Dataset | 99.69%; Precision: 98.27%, Recall: 98.26% | Transfer Learning, Explainability |
| IncepV3Dense for Banana [21] | Identify micronutrient deficiency in banana | InceptionV3 + DenseNet Ensemble | Mendeley Banana Crop | 98.62%; F1-score: 93% | Deep Ensemble Learning |
| Deep CNN for Detection [27] | End-to-end CNN for plant disease recognition | 8-layer Custom CNN | 147,500 mixed images | 99.97%; Precision/Recall/F1: ∼99.8% | Full CNN Training Pipeline |
| PlantDet Ensemble [32] | Multi-model ensemble for robustness | CNN + VGG + ResNet Ensemble | Rice, Betel Leaf | 98.53%; Precision: 98.5%, Recall: 98.35%, F1: 98.42% | Ensemble Strategy, TL |
| Wheat Leaf Classification [29] | Improve limited dataset learning via augmentation | MobileNetV2 | Balanced Wheat Dataset | 100%; Not Specified | CycleGAN, ADASYN, Augmentation |
| EffNet + GWT [12] | Improve cross-crop detection accuracy | EfficientNet + Group-wise Transformer | PV, Cassava, Tomato | 99.8%, 86.9%, 99.4%; Not Specified | Transformer Attention, TL |
| LeafNet Groundnut [28] | Boost accuracy with residual learning | LeafNet + Residual Connections | Groundnut (10,361 images) | 97.225%; Precision/Recall/F1: ∼97.2% | Weight Init, Residual Blocks |
| FL-ToLeD (Tomato) [4] | Lightweight detection for mobile devices | FL-ToLeD + Attention | Tomato (Bangladesh) | 99.04%; Precision/Recall/F1: 99% | Attention, Efficiency |
| CNN + ViT Hybrid [1] | Combine CNN features with transformer | CNN + Vision Transformer | Apple, Corn | 99.24%, 98%; Not Specified | Hybrid CNN-ViT Features |
| Segmented Detection [11] | Use segmentation before classification | SegNet + Classifier | Tomato (PV-derived) | 99.95%, 99.12–99.89%; Not Specified | Segmentation + CNN |
| Pretrained vs Custom CNN [3] | Benchmark pre-trained vs custom CNNs | Custom CNN + Others | Custom Leaf Dataset | Competitive; Not Specified | Model Evaluation Study |
| GSMo-CNN Multi-output [37] | Simultaneous disease + plant identification | GSMo-CNN | PV, PlantDoc, Plant Leaves | Up to 99.74%; Not Specified | Attention, Multi-Output |
| EffNet for Maize [17] | Efficient classification of maize diseases | EfficientNet (Fine-tuned) | Maize (8 classes) | 98.52%; Not Specified | TL, Compound Scaling |
| **Hybrid Model (Ours)** | Combine customized and pre-trained features for robust classification | Customized CNN + EfficientNetB0 | PlantVillage, New Plant Diseases (balanced) | **pv: 99.82%, 99.82%, and 99.82%; npd: 99.87%, 99.88%, and 99.87%** | Feature fusion, early stopping, basic augmentation |

As summarized in Table 6.7, the evaluated models align with or exceed the performance of many prior studies. For example, CNN-based models have previously achieved near-perfect accuracy using curated datasets [27], while ensemble methods and transfer learning strategies have reported accuracy in the 98–99% range [22, 32, 12]. Compared to these benchmarks, the EfficientNetB0 and Hybrid models in this study provide competitive or improved performance with a more streamlined training pipeline.

These results emphasize that modern architectures like EfficientNet can achieve excellent results even without sophisticated ensemble strategies or overly complex augmentation. Moreover, the results validate previous claims that dataset balance plays a critical role in reducing misclassifications and improving generalization [29, 10].

# Chapter 7

# Conclusion and Future Scope

This study conducted a comparative evaluation of three deep learning approaches—customized CNN, Pre-trained EfficientNetB0, and a Hybrid Model—for multiclass crop disease classification using two balanced datasets: PlantVillage and New Plant Diseases.

The Pre-trained EfficientNetB0 model delivered the highest overall performance, especially on the New Plant Diseases dataset, achieving the top accuracy (99.89%) along with superior precision, recall, and F1-score. The Hybrid Model closely followed and demonstrated robust generalization with minimal augmentation, achieving 99.87% accuracy. On the PlantVillage dataset, all models performed exceptionally well, with EfficientNetB0 and the Hybrid Model both achieving near-identical metrics.

Although effective, the customized CNN lagged slightly behind in performance, particularly on complex datasets. However, it remains a viable lightweight alternative for constrained environments [4, 25].

These findings suggest that while hybrid architectures offer performance benefits in specific settings [1, 5], fine-tuned pre-trained models like EfficientNetB0 remain highly competitive, especially in resource-efficient applications [12, 17]. Dataset balancing and simple augmentation techniques contributed significantly to all models' performance, reaffirming their importance in real-world deployment scenarios [29, 10].

The current study can be extended in multiple promising directions to enhance its real-world applicability and scientific depth. One critical avenue is the validation of deep learning models in uncontrolled farm environments, accounting for real-world variabilities such as inconsistent lighting, diverse camera qualities, and different crop growth stages. Additionally, compressing the hybrid model into a lightweight version will enable deployment on edge devices like smartphones, making disease diagnosis more accessible to farmers in remote areas. Integrating explainable AI techniques such as Grad-CAM or LIME can enhance transparency, allowing agronomists and farmers to interpret model decisions and build trust in automated systems. Future work could also involve expanding the classification capability to include disease severity levels—distinguishing between mild, moderate, and severe infections—to support more nuanced intervention strategies. Continuous learning systems can be developed to allow the model to retrain periodically using new data, thereby adapting to novel diseases or environmental changes. Lastly, incorporating multimodal data such as temperature, humidity, and soil conditions can significantly improve the robustness and reliability of predictions, enabling a more holistic approach to precision agriculture.

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